D02 · Demand Planning & Forecasting

05 · CPFR & collaborative demand planning (VMI)

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1 WHAT IT IS

CPFR and VMI replace independent dual forecasting with a shared, reconciled demand plan across the trading-partner boundary — the party closest to the end signal owns the forecast, and replenishment runs off one number instead of two distorted ones.

2 WHY IT MATTERS

Aviv 2001 **(Aviv, 2001 — peer-reviewed)** showed that collaborative forecasting captures value precisely when partners hold different, complementary information about the same demand — and that the gain comes from sharing the signal, not from a better algorithm on either side alone. When supplier and customer each forecast the same end demand against different signals — the supplier sees orders, the customer sees POS — they institutionalize the bullwhip and both carry safety stock against the same uncertainty. Duplicate buffering across the boundary commonly adds **20–35% to total channel inventory** for no service gain, because each party is hedging a number the other party distorted.

3 HOW IT IS DONE

1. **Establish the single shared signal.** Agree which data — POS, on-hand, sell-through — is the one source of truth, and who owns it. Two partners cannot collaborate on a forecast until they agree what they are forecasting against.
2. **Assign forecast ownership by proximity to the end signal.** For POS-driven categories that is usually the retailer; for engineered components it may be the supplier. The owner forecasts; the partner contributes information, not a competing number.
3. **Reconcile, do not average.** A joint forecast process surfaces and resolves the gap between the two views. Averaging two forecasts hides the disagreement and the information behind it — the gap is the signal, not the noise.
4. **Collapse the duplicate buffers.** With one shared signal, size a single channel safety stock against the true end-demand variance — not two buffers against the laundered order stream. This is where the inventory reduction is realized.
5. **Define exception handling jointly, in advance.** Agree the thresholds and the escalation path before the disruption, not during it. CPFR's value shows up most when demand surprises — and only if the exception process was built when both parties were calm.
6. **Measure at the channel level.** Track total channel inventory, channel fill rate, and forecast error at the shared signal — not each firm's local optimum. Two locally optimized inventories are usually one globally bloated channel.

> **WORKED EXAMPLE — ILLUSTRATIVE**
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> A supplier and a major retailer each forecast the same 600-SKU line independently — **baseline: combined channel inventory $52M, channel fill 95%, both parties buffering against the same uncertainty**. They implemented CPFR on shared POS with a retailer-owned forecast and joint exception handling. Results: **channel inventory $52M -> $40M (-23%), channel fill 95% -> 97%, forecast error at the shared signal -19%** — with bullwhip variance amplification across the boundary cut roughly in half.

IN PRACTICE

Single shared signal governance

Define the shared data, its definition, its refresh cadence, and its owner in writing. Most CPFR programs fail not on algorithms but on data — disagreement over what "sell-through" means, or a feed that lags, quietly recreates the two-number problem the program was meant to solve.

Reconcile-not-average discipline

When the two views diverge, hold a structured reconciliation that asks *why* — the supplier sees a capacity constraint, the retailer sees a planned promotion. Resolving the gap surfaces information; splitting the difference buries it. Build the cadence and the forum, not just the spreadsheet.

Joint exception thresholds

Pre-agree the deviation thresholds that trigger a joint review and the escalation path when they breach. The VICS CPFR model is built around exception management — collaboration on the 95% that goes to plan is cheap; the value is in a pre-built process for the 5% that does not.

Channel-level metrics over local optima

Measure and reward total channel inventory and channel service, not each firm's local balance sheet. Without channel-level metrics, each party optimizes locally and the duplicate buffer creeps back — the scorecard is what holds the collaboration in place.

WHAT YOU WOULD USE

CPFR / COLLABORATIVE PLANNING

SAP IBP Response & Supply / o9 control tower — enterprise tier-1 — Native multi-enterprise collaboration on a shared demand and supply plan, with exception-based workflow and a common data model. **(SAP, o9 — vendor)** The choice when collaboration must extend across the full planning stack of both partners.

Blue Yonder / Kinaxis collaborative planning — Mature collaborative demand and replenishment with strong VMI and exception management. **(Blue Yonder, Kinaxis — vendor)** A strong fit where the supplier runs replenishment on behalf of the customer and needs a shared workspace.

Retailer-supplier collaboration portals / VMI specialists — Dedicated collaboration and VMI platforms that connect a supplier to many retail partners. **(specialized VMI vendors — vendor)** The vertical option for suppliers managing collaborative replenishment across a fragmented retail base.

For operations that don't need a platform — A shared scorecard plus an EDI 852 (product activity / POS) feed and a recurring joint review on a fixed cadence. The single shared signal and the joint exception process — not the platform — are what dissolve the duplicate buffer; the tooling can follow once the discipline holds.

4 SO WHAT

Operations that plan off the shared downstream signal carry one buffer against the real uncertainty; operations that forecast against each other's orders carry two buffers against a number they distorted together. CPFR is not a relationship exercise — it is the structural removal of a buffer that exists only because two parties refused to share one signal. The duplicate inventory across the boundary is the price of that refusal, paid every cycle.

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